

Lab 4 - Python Assignment

Question 1

Write a Python program to find the smallest number in the list `my_list = [3, 1, 0, 9, 5, 2, 6, 4, 9, 8, 7]` using a for loop. Print the smallest number as the output.

Code:

```
my_list = [3, 1, 0, 9, 5, 2, 6, 4, 9, 8, 7]

smallest = my_list[0]
for num in my_list:
    if num < smallest:
        smallest = num

print("Smallest number:", smallest)
```

Output:

```
Smallest number: 0
```

Question 2

Write a Python program to find the average of all the numbers in `my_list = [3, 1, 0, 9, 5, 2, 6, 4, 9, 8, 7]`. Use a for loop to calculate the sum and then find and print their average.

Code:

```
my_list = [3, 1, 0, 9, 5, 2, 6, 4, 9, 8, 7]

total = 0
for num in my_list:
    total += num

average = total / len(my_list)

print("Sum:", total)
print("Average:", average)
```

Output:

```
Sum: 54
Average: 4.909090909090909
```

Question 3

Create a list of student names: students = ["Ali", "Ahmed", "Sara", "Ayesha", "Bilal"]. Ask the user to enter a student name. If the name exists, display "Student Found", otherwise display "Student Not Found".

Code:

```
students = ["Ali", "Ahmed", "Sara", "Ayesha", "Bilal"]

name = input("Enter a student name: ")

if name in students:
    print("Student Found")
else:
    print("Student Not Found")
```

Output (name exists):

```
Enter a student name: Sara
Student Found
```

Output (name does not exist):

```
Enter a student name: Hamza
Student Not Found
```

Question 4

Create an empty list: shopping = []. Ask the user to enter 5 shopping items. Then display the complete shopping list.

Code:

```
shopping = []

for i in range(5):
    item = input("Enter shopping item " + str(i + 1) + ": ")
    shopping.append(item)

print("Shopping List:")
for item in shopping:
    print("-", item)
print(shopping)
```

Output:

```
Enter shopping item 1: Milk
Enter shopping item 2: Bread
Enter shopping item 3: Eggs
Enter shopping item 4: Rice
Enter shopping item 5: Apples
Shopping List:
- Milk
- Bread
- Eggs
- Rice
- Apples
['Milk', 'Bread', 'Eggs', 'Rice', 'Apples']
```